

EXPERIMENT 29:

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **Logistic -R 1.0E-8 -M -1 -num-decimal-places 4**

Test options:
☐ Use training set
☐ Supplied test set Set...
☒ Cross-validation Folds **10**
☐ Percentage split % **66**
More options...

(Nom) class
Start Stop

Result list (right-click for options):
21:06:53 - trees.J48
21:42:02 - trees.J48
21:42:45 - trees.J48
21:46:09 - functions.SMO
21:47:19 - meta.CostSensitiveClassifier
21:49:12 - functions.Logistic

Classifier output:

Time taken to build model: 0.12 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	752	75.2 %
Incorrectly Classified Instances	248	24.8 %
Kappa statistic	0.375	
Mean absolute error	0.3098	
Root mean squared error	0.4087	
Relative absolute error	73.727 %	
Root relative squared error	89.1751 %	
Total Number of Instances	1000	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.864	0.510	0.798	0.864	0.830	0.379	0.785	0.883	good
	0.490	0.136	0.607	0.490	0.542	0.379	0.785	0.599	bad
Weighted Avg.	0.752	0.398	0.741	0.752	0.744	0.379	0.785	0.798	

=== Confusion Matrix ===

```
a b <-- classified as
605 95 | a = good
153 147 | b = bad
```

Status: OK Log x 0

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **NaiveBayes**

Test options:
☐ Use training set
☐ Supplied test set Set...
☒ Cross-validation Folds **10**
☐ Percentage split % **66**
More options...

(Nom) class
Start Stop

Result list (right-click for options):
21:06:53 - trees.J48
21:42:02 - trees.J48
21:42:45 - trees.J48
21:46:09 - functions.SMO
21:47:19 - meta.CostSensitiveClassifier
21:49:12 - functions.Logistic
21:49:33 - bayes.NaiveBayes

Classifier output:

Time taken to build model: 0.01 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	754	75.4 %
Incorrectly Classified Instances	246	24.6 %
Kappa statistic	0.3813	
Mean absolute error	0.2936	
Root mean squared error	0.4201	
Relative absolute error	69.8801 %	
Root relative squared error	91.6718 %	
Total Number of Instances	1000	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.864	0.503	0.800	0.864	0.831	0.385	0.787	0.891	good
	0.497	0.136	0.611	0.497	0.548	0.385	0.787	0.577	bad
Weighted Avg.	0.754	0.393	0.743	0.754	0.746	0.385	0.787	0.797	

=== Confusion Matrix ===

```
a b <-- classified as
605 95 | a = good
151 149 | b = bad
```

Status: OK Log x 0

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **J48 -C 0.25 -M 2**

Test options:
☐ Use training set
☐ Supplied test set
☒ Cross-validation Folds **10**
☐ Percentage split % **66**

(Nom) class:

Result list (right-click for options):

- 21:06:53 - trees.J48
- 21:42:02 - trees.J48
- 21:42:45 - trees.J48
- 21:46:09 - functions.SMO
- 21:47:19 - meta.CostSensitiveClassifier
- 21:49:12 - functions.Logistic
- 21:49:33 - bayes.NaiveBayes
- 21:50:01 - trees.J48**

Classifier output:

Time taken to build model: 0.01 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	705	70.5	%
Incorrectly Classified Instances	295	29.5	%
Kappa statistic	0.2467		
Mean absolute error	0.3467		
Root mean squared error	0.4796		
Relative absolute error	82.5233 %		
Root relative squared error	104.6565 %		
Total Number of Instances	1000		

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
Weighted Avg.	0.705	0.475	0.687	0.705	0.692	0.251	0.639	0.657	

=== Confusion Matrix ===

```

a b <-- classified as
588 112 | a = good
183 117 | b = bad

```

Status: OK  x 0

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **libk -K 1 -W 0 -A "weka.core.neighboursearch.LinearNNSearch -A \"weka.core.EuclideanDistance -R first-last\""**

Test options:
☐ Use training set
☐ Supplied test set
☒ Cross-validation Folds **10**
☐ Percentage split % **66**

(Nom) class:

Result list (right-click for options):

- 21:06:53 - trees.J48
- 21:42:02 - trees.J48
- 21:42:45 - trees.J48
- 21:46:09 - functions.SMO
- 21:47:19 - meta.CostSensitiveClassifier
- 21:49:12 - functions.Logistic
- 21:49:33 - bayes.NaiveBayes
- 21:50:01 - trees.J48
- 21:50:26 - lazy.libk**

Classifier output:

Time taken to build model: 0 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	720	72	%
Incorrectly Classified Instances	280	28	%
Kappa statistic	0.3243		
Mean absolute error	0.2805		
Root mean squared error	0.5286		
Relative absolute error	66.7546 %		
Root relative squared error	115.3422 %		
Total Number of Instances	1000		

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
Weighted Avg.	0.720	0.400	0.716	0.720	0.718	0.325	0.660	0.669	

=== Confusion Matrix ===

```

a b <-- classified as
567 133 | a = good
147 153 | b = bad

```

Status: OK  x 0

Weka Explorer

PreprocessClassifyClusterAssociateSelect attributesVisualize

Classifier

ChooseSMO -C 1.0 -L 0.001 -P 1.0E-12 -N 0 -V -1 -W 1 -K "weka.classifiers.functions.supportVector.PolyKernel -E 1.0 -C 250007" -calibrator "weka.classifiers.functions.Logistic -R 1.0E-8 -M -1 -num-decimal-places 4"

Test options

☐ Use training set

☐ Supplied test set

Set...

☒ Cross-validation

Folds10

☐ Percentage split

%66

More options...

(Nom) class

Start

Stop

Result list (right-click for options)

21:06:53 - trees.J48

21:42:02 - trees.J48

21:42:45 - trees.J48

21:46:09 - functions.SMO

21:47:19 - meta.CostSensitiveClassifier

21:49:12 - functions.Logistic

21:49:33 - bayes.NaiveBayes

21:50:01 - trees.J48

21:50:26 - lazy.IBk

21:50:50 - functions.SMO

Classifier output

Time taken to build model: 0.22 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	751	75.1	%
Incorrectly Classified Instances	249	24.9	%
Kappa statistic	0.3654		
Mean absolute error	0.249		
Root mean squared error	0.499		
Relative absolute error	59.2607	%	
Root relative squared error	108.8905	%	
Total Number of Instances	1000		

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.871	0.530	0.793	0.871	0.830	0.371	0.671	0.781	good
	0.470	0.129	0.610	0.470	0.531	0.371	0.671	0.446	bad
Weighted Avg.	0.751	0.410	0.738	0.751	0.741	0.371	0.671	0.681	

=== Confusion Matrix ===

a

b

<-- classified as

610	90		a = good
159	141		b = bad

Status

OK

Log

x 0